Ozlem Yıldız

New York City, USA · (+1) 646-217-8249 · ozlemyildizee@gmail.com
Website · LinkedIn · Google Scholar · GitHub

Summary

Research scientist working on learning sharing between coding agents. Prior work spans agent evaluation and data, multimodal LLM post-training, automated data curation, and LLM-as-a-judge reward systems shipped in Ray-Ban Meta AI Glasses across 11+ languages. Ph.D. in ECE, NYU Tandon. Two patents pending, eight publications.

Technical Skills

ML & AI: PyTorch, HuggingFace Transformers, RLHF, DPO, SFT, Reward Modeling, LLM Post-training, AI Agents & Agent Evaluation, Benchmark Design, LLM-as-a-Judge Evaluation, Multimodal Models (text, vision, audio), Prompt Engineering, Computer Vision, TensorFlow

Systems & Infra: Python, SQL, Async/Concurrent Systems, Distributed Training (`torch.distributed`), CUDA, HPC/GPU Clusters, Workflow Orchestration, Distributed Storage, Linux, C/C++, Java

Research Tools: MATLAB, Simulink, 5G NR Toolbox, Ray Tracing Toolbox

Work Experience

Meta *New York, NY*

Research Scientist, Agent Data & Optimization *May 2026 – Present*

- Built a plugin for an AI coding agent harness that shares learned context across agents, preventing repeated failures on previously-solved problems
- Designed benchmark tasks and evaluation frameworks to measure and improve AI coding agents, defining what "good" means for agentic coding behavior and translating it into reproducible, measurable criteria

Research Scientist, Multimodal AI, Ray-Ban Meta Glasses *September 2025 – May 2026*

- Built a production translation evaluation and auto-rewrite workflow with iterative improvement loops, raising success rates from 48% to 90% (+42pp) across European languages
- Drove the Turkish voice LLM launch on Ray-Ban glasses, lifting LLM-judge translation quality score from 67.8 to 86.4 (+18.6) over 1M samples including phonetic error cases encountered in production; judge calibrated against human annotations and reused as the RL reward signal
- Post-trained multimodal models for intelligent AR wearables, improving image-based translation quality by 14.7% and contributing to AI task success rate gains on text-in-image queries
- Architected and shipped a fully async data curation pipeline that replaced manual per-topic workflows, producing 100K+ quality-checked training samples across 11 languages, with adaptive concurrency control, buffered checkpoint writes, and per-phase fault isolation for crash-resilient resume
- Designed the pipeline's multi-dimensional LLM-as-a-judge framework spanning topic relevance, image quality, redaction detection, translation grading, refusal detection, and helpfulness

Meta (Internship) *Burlingame, CA*

PhD Software Engineer Intern, AR/VR *June – August 2024*

Supervised by Dr. Margarita Grinvald

- Independently delivered an end-to-end ML inpainting model for depth and texture synthesis at disocclusion regions arising from camera-to-eye reprojection in AR headsets, operating from local context

- Integrated a layered depth inpainting model into a production AR/VR passthrough rendering pipeline; built camera model projections, coordinate transforms, temporal stability via blending, and GPU kernel testing for real-time rendering across 100+ frame sequences
- Achieved measurable improvements in reconstruction quality (PSNR/SSIM) over baselines, directly enhancing visual fidelity in the AR display pipeline

NYU Wireless *New York, NY*
Research Assistant *August 2020 – August 2025*
Advisor: Prof. Elza Erkip

- Designed deep neural networks for nonlinear channel capacity estimation under hardware constraints (with Prof. Sundeep Rangan)
- Developed a group-testing framework for analog/hybrid beam alignment to detect multiple paths in mmWave channels
- Built neural network architectures for task-based and semantic joint source-channel coding (with Prof. Yao Wang)
- Investigated VR streaming quality over wireless channels; aligned neural cellular automata with pathfinding problems using learned and hand-coded models

Samsung Research America *Dallas, TX*
Graduate Research Intern *June – August 2023*
Supervised by Dr. Jianhua Mo & Dr. Ahmad AlAmmouri

- Designed ML-based optimization of 3D frequency-dependent beamforming using true time delay elements and phase shifters for next-generation wireless systems
- Received **Best Poster Award** among all intern final presentations
- Work led to a publication at IEEE VTC 2024 and a U.S. patent application

Dell Technologies *New York, NY*
Graduate Research Intern *June – August 2022*
Supervised by Dr. Tejinder Singh

- ML-based optimization of secrecy capacity in IRS-assisted mmWave indoor wireless systems; contributed to a U.S. patent application

InterDigital *New York, NY*
R&I Intern *June – August 2021*
Supervised by Dr. Ramon Khalona

- Evaluated non-linear waveform spectral performance and developed a waveform energy KPI for high-frequency wireless communications

Education

Ph.D., Electrical and Computer Engineering *GPA: 3.93 / 4.0*
NYU Tandon School of Engineering, New York, USA *2020 – 2025*

- Advisor: Prof. Elza Erkip, School of Engineering Fellowship
- Thesis: *Hardware-Aware Design and Optimization for Millimeter Wave and Sub-Terahertz Wireless Systems*
- Teaching: Probability & Stochastic Processes (Prof. Erkip), Machine Learning (Prof. Musco)

B.Sc., Electrical and Electronics Engineering *GPA: 3.83 / 4.0*
Bilkent University, Ankara, Turkey *2015 – 2020*

- Fellowship student (ranked **215 / 2 million** in national university placement exam)
- Erasmus+ Exchange: University of Erlangen-Nuremberg, Germany (March – August 2019)

Publications

P. Yang, **O. Yıldız**, R. Nikandish, W. Xia, S. Rangan, E. Erkip, H. Rahmani, “Hardware-Algorithm Co-Design for Interference Mitigation in FR3 Receivers,” *accepted at Asilomar 2026*.

O. Yıldız, S. Rangan, H. Rahmani, E. Erkip, “Hardware Constrained Receiver Under Strong Structured Interference,” *submitted to IEEE Transactions on Wireless Communications*.

O. Yıldız, A. AlAmmouri, J. Mo, Y. Nam, E. Erkip, J. Zhang, “3D Beamforming Through Joint Phase-Time Arrays,” *IEEE Vehicular Technology Conference (VTC)*, 2024.

O. Yıldız, M. Alavirad, T. Singh, “Investigation and Optimization of Secrecy Capacity for Intelligent Reflective Surfaces-Assisted Secure mmWave Indoor Wireless Communication,” *IEEE Radio and Wireless Symposium (RWS)*, 2023.

S. Earle, **O. Yıldız**, J. Togelius, C. Hegde, “Pathfinding Neural Cellular Automata,” *arXiv:2301.06820*.

F. Wilhelmi, J. Hribar, S. F. Yilmaz, E. Ozfatura, K. Ozfatura, **O. Yıldız**, D. Gündüz, H. Chen, X. Ye, L. You, Y. Shao, P. Dini, B. Bellalta, “Federated Spatial Reuse Optimization in Next-Generation Decentralized IEEE 802.11 WLANs,” *ITU Journal on Future and Evolving Technologies (ITU J-FET)*, 2022.

O. Yıldız, A. Khalili, E. Erkip, “Hybrid Beam Alignment for Multi-path Channels: A Group Testing Viewpoint,” *IEEE Asilomar Conference on Signals, Systems, and Computers*, 2022.

E. Erturk, **O. Yıldız**, S. Shahsavari, N. Akar, “Power Allocation and Temporal Fair User Group Scheduling for Downlink NOMA,” *Telecommunication Systems*, vol. 77, pp. 753–766, 2021.

Patents

O. Yıldız, A. AlAmmouri, J. Mo, Y. Nam, “3D Beamforming Through Joint Phase-Time Arrays,” U.S. Patent App. 18/811,643, filed August 2024 (pending).

O. Yıldız, M. Alavirad, T. Singh, “Increasing Secrecy Capacity for Intelligent Reflective Surface-Assisted Wireless Communications,” U.S. Patent App. 18/469,724, filed September 2023 (pending).